

Ziad Ahmed Rabie

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Summary

Computer Engineering student with strong foundations in software engineering, mathematics, and algorithms. Focused on applied AI/ML, with practical experience in computer vision, deep learning pipelines, and data-driven modeling. Interested in building intelligent systems that sit at the intersection of machine learning and real-world software.

Education

Egypt University of Informatics (EUI) Oct 2023 – Present
B.Sc. Computer Engineering CGPA: 3.96 / 4.00
• Full scholarship awarded to top-ranked Thanaweya Amma students in Egypt.

Technical Skills

Languages: Python, C/C++, JavaScript, TypeScript, HTML/CSS, MATLAB
AI / ML: PyTorch, Scikit-learn, OpenCV, NumPy, Pandas, SymPy
Web & Software: React, Next.js, FastAPI, PySide6
Embedded Systems: Embedded C, AVR Microcontrollers, Arduino
Tools: Git/GitHub, Power BI

Projects

MathGPT *Team Project*
Python, React, PyTorch, OpenCV, SymPy, Ollama, CUDA

- Built an end-to-end handwritten math solver integrating CNN-based symbol recognition, OpenCV segmentation, SymPy computation, and LLM-generated explanations.
- Contributed to model training, segmentation pipeline, and full-stack integration; achieved **>97% validation accuracy** on symbol classification.

AI Portfolio Optimization System *Team Project*
Python, FastAPI, Next.js, Scikit-learn, XGBoost, Power BI

- Developed ML-driven workflows for return forecasting, risk modeling, and strategy backtesting.
- Contributed to model development, Power BI dashboards, and data reporting for portfolio decision support.

Warsha Hub *Team Project*
Python, PySide6, C++, SQLite

- Built a graph-based social platform with network visualization, recommendations, and pathfinding.

Electronic Organ *Team Project*
Circuit Design, Embedded Logic

- Designed an audio oscillator for tone generation using analog circuit design and embedded logic.

Experience

NAID Computer Vision Internship Summer 2025
AI, Data Analysis & Assistive Technologies (64 hrs)

- Explored AI, data analysis, and computer vision methodologies to design accessibility applications.
- Contributed to **InSync**, an assistive app integrating image segmentation, audio denoising, and text-to-speech.

ITI Embedded Systems AVR Program Summer 2025
Embedded C, AVR Microcontrollers, Hardware Interfacing (162 hrs)

- Completed intensive training in AVR architecture, memory-mapped I/O, and peripheral interfacing.
- Developed a multi-function digital clock with RTC, alarm, stopwatch, and countdown timer on AVR.

Leadership & Activities

- **iMath Club — Co-Founder (EUI):** Managed and organized math competitions and initiatives.
- **ICPC Club — Mentor (EUI):** Mentored peers in algorithms, data structures, and competitive programming.
- **Math & Physics Tutor:** Provided university-level tutoring with emphasis on analytical problem-solving.
- **DSC MENA 2024 — Organizer:** Coordinated logistics for a regional AI and Data Science conference.
- **IEEE EUI — Social Media Manager:** Produced technical content and managed student branch outreach.

Achievements

- **14th Place** — ICPC ECPC Qualifications (2024 & 2025)
- Top achiever at EUI (CGPA 3.96/4.00) & top-ranked in Thanaweya Amma national examinations
- **1st Place** — KSA Globe Program (Middle School)